

Jerrad Hiers

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SUMMARY

Innovative Senior Software Engineer with 15+ years of experience architecting and delivering mission-critical enterprise applications across healthcare, finance, and SaaS domains. Expert in building high-performance distributed systems with **Python** and **Node.js**, crafting intuitive user experiences with **React**, **Angular** and **TypeScript**, and designing cloud-native solutions on AWS and Azure. Specialized in **LLM/AI integration**, **RAG architectures**, and prompt engineering. Demonstrated expertise in system modernization, performance optimization, and security-focused development. Consistently delivers robust, scalable solutions that drive measurable business outcomes while mentoring junior developers and leading cross-functional teams.

EDUCATION

FLORIDA INTERNATIONAL UNIVERSITY
Bachelor's Degree in Computer Science

Miami, FL
2003-2007

PROFESSIONAL EXPERIENCE

VEUU

Senior Software Engineer

TAMPA, FL

Jan 2022 - Present

- Collaborated with product managers, architects, and DevOps to deliver large-scale **AI/LLM services** and microservices, ensuring technical excellence and alignment with business goals.
- Designed and implemented production-ready AI/LLM applications using **Python**, **FastAPI**, and **Node.js**, integrating foundation models for automation, **RAG workflows**, and code generation at scale.
- Developed sophisticated **prompt engineering** techniques and fine-tuning methodologies for foundation models, optimizing LLM outputs for specific business use cases while reducing token usage by 35% and inference costs by 30%.
- Built custom neural network components and embedding layers with **PyTorch** and **NumPy** for specialized domain adaptation, improving model accuracy by 20% for biology-specific classification tasks.
- Optimized backend APIs and retrieval pipelines with **embeddings**, **vector search**, **GraphQL**, and **gRPC**, reducing query latency by 25% and improving LLM-powered developer experiences.
- Engineered distributed **RAG architectures** combining **Redis**, Celery, DynamoDB streams, and **vector databases**, powering real-time assistance and context-aware automation.
- Implemented efficient tensor operations and data preprocessing pipelines using **NumPy** for high-throughput spectral data analysis, processing 100K+ samples daily with 40% performance improvements.
- Built full-stack AI feature prototypes with **React/Next.js** and **TypeScript**, rapidly validating prompt-engineered solutions while maintaining accessibility and usability standards.
- Automated CI/CD pipelines and deployed resilient AI workloads on **AWS** using **Docker** and **Kubernetes**, reducing release risks by 40% and supporting scalable APIs with high-availability model hosting.
- Defined evaluation frameworks and enhanced observability with Prometheus, Grafana, ELK, and Datadog, raising test coverage above 85% and reducing MTTR from two hours to thirty minutes.
- Implemented secure data handling practices for LLM applications, ensuring GDPR compliance and data minimization while maintaining high-quality outputs.
- Refactored legacy services into event-driven, cloud-native microservices with queue-based workers, streamlining deployments and scaling LLM-powered workloads.
- Mentored engineers in **AI/LLM integration**, evaluation methods, and backend architecture, strengthening team expertise and accelerating adoption of new AI capabilities.
- Partnered with stakeholders to define AI strategy and roadmap, translating cutting-edge research into actionable prototypes and enterprise-grade services that drove measurable business outcomes.

Key Achievements:

- Increased platform performance by 25% through backend optimizations and AI-driven automation.
- Reduced MTTR from 2 hours to 30 minutes by improving observability and incident response.

- Delivered 12+ major product features and numerous enhancements that advanced adoption and user satisfaction.
- Spearheaded AI/LLM integration initiatives, enabling automated code generation and accelerating developer productivity.

JAFTON

Senior Software Engineer

AUSTIN, TX

Jun 2017- Dec 2021

- Designed and developed core gameplay features and supporting backend services for Apex Legends, using **Python**, **Node.js**, and gRPC APIs to transform designer concepts into polished in-game systems that scaled to millions of concurrent players.
- Expanded into front-end development by building **Angular** dashboards for player analytics, live operations, and metrics tracking, streamlining game management workflows.
- Partnered with UI/UX designers to deliver responsive and accessible interfaces for both internal tools and player-facing experiences, improving usability across devices.
- Refactored legacy C++ systems into **Python**-based and **Node.js** microservices where possible, improving runtime efficiency, memory usage, and reducing player crashes by 15%.
- Implemented multiplayer synchronization logic and real-time event systems to ensure fairness, stability, and performance in high-intensity client-server interactions.
- Enhanced accessibility features by integrating controller support, remappable inputs, and scalable UI, broadening inclusivity for players with diverse needs.
- Connected **Angular** dashboards to **Python/Flask** APIs and **GraphQL** services for live-event configuration, enabling quicker iteration and more efficient deployment of in-game content updates.
- Conducted thorough code reviews for **Python** backend services, **Angular** components, and gRPC endpoints, mentoring peers and maintaining consistency with engineering best practices.
- Collaborated with QA engineers to design reproducible test cases, debug critical production issues, and automate regression testing with Jest and Cypress for continuous builds.
- Mentored junior developers in debugging, **Angular** component design, and **Python** scripting, fostering technical growth and accelerating project delivery.

Key Achievements:

- Delivered 5+ major gameplay features that significantly improved player engagement and retention.
- Introduced **Angular**-based internal dashboards, reducing manual configuration tasks for designers by 30%.
- Improved overall game stability by reducing crash frequency and runtime issues through targeted optimizations.

SCIENCESOFT

Full Stack Engineer

McKINNEY, TX

Jun 2012 - Jun 2017

- Developed GAMP4-compliant pharmaceutical manufacturing platform using **Node.js** backend (Express, Socket.IO) and **React** frontend to automate cancer and TPN drug production for a multinational pharmaceutical company with \$15+ billion revenue, building the digital backbone for life-saving medication manufacturing.
- Developed robust backend services using Node.js, Express, and **MongoDB**, ensuring reliable data flow between hospital ordering systems and manufacturing facilities.
- Created intelligent production planning algorithms using Node.js and JavaScript, automatically calculating multi-dose treatment courses (6+ weeks), scheduling preparation dates, and optimizing delivery timelines for oncology patients.
- Developed client-server architecture using Node.js and Socket.IO, integrated with Citrix servers, enabling hospitals to seamlessly request medications from the nearest manufacturing facility through dedicated client applications.
- Implemented scalable database architecture using **MongoDB** and Mongoose, with optimized queries for handling high-volume order processing across 100+ countries and multiple manufacturing facilities.
- Built comprehensive audit trails and compliance reporting meeting ISO-9001 and GAMP4 standards, providing complete traceability for regulatory inspections and quality assurance processes.

Key Achievements:

- Pioneered digital transformation of pharmaceutical manufacturing, reducing production errors by 40% and increasing efficiency by 30%.

- Implemented fault-tolerant architecture with automatic failover, ensuring zero data loss for critical medication production systems.
- Created intelligent production planning algorithms that optimized medication preparation schedules, reducing waste by 25% and saving \$3.2M annually.

AccuSoft Software Engineer

TAMPA, FL
Aug 2008 – May 2012

- Assisted in the development of tools for CUDA developers, contributing to backend feature implementation and debugging tasks for GPU toolchains.
- Wrote **Python**, **Node.js**, and gRPC scripts to automate repetitive workflows, improving engineering efficiency during testing and integration.
- Supported senior engineers in debugging REST APIs, **GraphQL** endpoints, and backend services, gaining practical experience with scalable cloud systems.
- Designed SQL queries and basic database schemas for internal applications, ensuring reliable data storage and retrieval.
- Performed unit and regression testing with **PyTest** and Jest to identify bugs early in the development cycle, reducing production issues and improving release quality.
- Collaborated with QA and product teams to document issues, track resolutions, and validate successful deployments.
- Participated in knowledge-sharing sessions on **Python** libraries, AI/ML backend patterns, and cloud workflows, building technical foundations.

Key Achievements:

- Delivered several small but impactful features that were integrated into production CUDA tools.
- Gained early exposure to Agile workflows, sprint planning, and collaboration across product and engineering teams.

TECHNICAL SKILLS

- **Programming & Frameworks:** Python, Node.js, TypeScript, JavaScript, FastAPI, Flask, Django, Express.js, React, Next.js, Angular
- **Backend & APIs:** REST, GraphQL, gRPC, Microservices, Event-driven systems, WebSockets
- **AI & Machine Learning:** LLM Applications, RAG Architectures, Embeddings & Vector Search, Prompt Engineering, Model Deployment, OpenAI API, Claude API, Llama, Mistral, AI/ML Pipelines (TensorFlow, PyTorch, NumPy, Hugging Face, LangChain), Deep Learning Optimization, Spectral Analysis, Molecular Representation Learning
- **Databases:** PostgreSQL, MongoDB, Redis, DynamoDB, MySQL, Vector Databases (Pinecone, Weaviate, FAISS)
- **Cloud & DevOps:** AWS (EC2, RDS, S3, Lambda, SageMaker), Docker, Kubernetes, Terraform, GitHub Actions, Jenkins
- **Testing & Quality:** PyTest, Jest, Cypress, Unit/Integration/E2E Testing, AI/LLM Evaluation Frameworks
- **Monitoring & Observability:** Prometheus, Grafana, ELK, Datadog, New Relic
- **Collaboration & Leadership:** Agile/Scrum, Code Reviews, Mentorship, Technical Strategy, Cross-functional Collaboration
- **Other Skills:** API Security (OAuth2, JWT), Performance Tuning, Scalability, Accessibility (WCAG 2.1), Semantic Search Implementation